

An effective hybrid genetic algorithm for the multi-robot task allocation problem with limited span

Wenbo Liu ^{a,b}, Zhian Kuang ^c, Yongcong Zhang ^{a,b}, Bo Zhou ^{a,b}, Pengfei He ^{a,b}, Shihua Li ^{a,b}

^a School of Automation, Southeast University, Nanjing, China

^b Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, Nanjing, China

^c Shanghai Friendless Electronic Technology, Shanghai, 201100, PR China

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ABSTRACT

Multi-robot task allocation is one of the most interesting multi-robot systems that have gained considerable attention due to various real-world applications. In this paper, we focus on a multi-robot task allocation problem where a set of industrial robots, which are installed on a gantry and have a limited working span, have to jointly perform a set of weld lines in large workpieces. Considering the emphasis on minimizing the processing time of workpieces in industry, the objective of this problem is to minimize the cycle time when scheduling a set of robots to work together efficiently. Following practical applications, we present a mathematical model for small size instances, and for large size instances, we propose an effective hybrid genetic algorithm to solve it because of the significant computational complexity, which includes a specific region division method is used to divide the workpieces into a set of regions where the robots can reach all the weld lines in each region, a dedicated route-based crossover to generate promising offspring solutions, and an effective neighborhood-based local search procedure to improve each offspring solution as much as possible. Extensive experimental results on three benchmark instances show that the algorithm significantly outperforms two refer methods with an average improvement of 6.06% and 4.6%. Additional experiments on real-world instances are presented to verify the algorithm's effectiveness in solving the multi-robot task allocation problem with limited span.

1. Introduction

In the industrial landscape, the integration of multi-robot systems has achieved widespread adoption (Qin et al., 2023; Verma & Ranga, 2021), demonstrating versatility across diverse industrial sectors, including but not limited to welding (Guo et al., 2024; Rout et al., 2021), assembly (Eswaran et al., 2024), disassembly (Wu et al., 2023), and stacking (Ko & Han, 2022). Within the framework, the multi-robot task allocation problem (MRTA) emerges as a cornerstone problem, delineating the optimization of resource allocation and task assignment among industrial robots (Murugappan et al., 2022; Qin et al., 2024), thereby engendering heightened operational efficiency and cost-effectiveness. The MRTA is customarily calibrated to encompass generic scenarios wherein multiple robots perform machining operations on workpieces (Yan et al., 2024). Notably, in Fig. 1, in some assembly line configurations, multi-robots are often installed on mobile platforms in order to increase the robot's operational flexibility and working range. As shown in Fig. 2(a), two robots (manipulators) are installed on a

movable gantry, resulting in each manipulator being associated with a limited span when the position of gantry is given. As a result, the task allocation of two robots becomes a challenging problem, because the working span of each robot should also be considered, which makes the problem more complex to solve. In this paper, we formulate the problem as a new variant of MRTA: the MRTA with limited span (MRTALS), where the working span of each robot is influenced by the position of a movable gantry.

The MRTA typically has a specific objective function in real-world applications, such as allocating tasks to robots to minimize cycle time. Notably, Gerkey and Mataric (2004) presented a comprehensive taxonomy of the problem, delineating it based on robots' capabilities, task requirements, and temporal considerations. Subsequently, Korsah et al. (2013) augmented the taxonomy by incorporating interrelated utilities (costs) and task constraints, thus elucidating the intricate interplay between robot-task costs. According to the taxonomy, the MRTALS presents two distinct dependencies: (i) cross-schedule dependencies,

* Corresponding author at: School of Automation, Southeast University, Nanjing, China.

E-mail addresses: lwb@seu.edu.cn (W. Liu), kuangzhian@fscut.com (Z. Kuang), zycongzz426@gmail.com (Y. Zhang), zhoubo@seu.edu.cn (B. Zhou), pengfeihe606@gmail.com (P. He), lsh@seu.edu.cn (S. Li).

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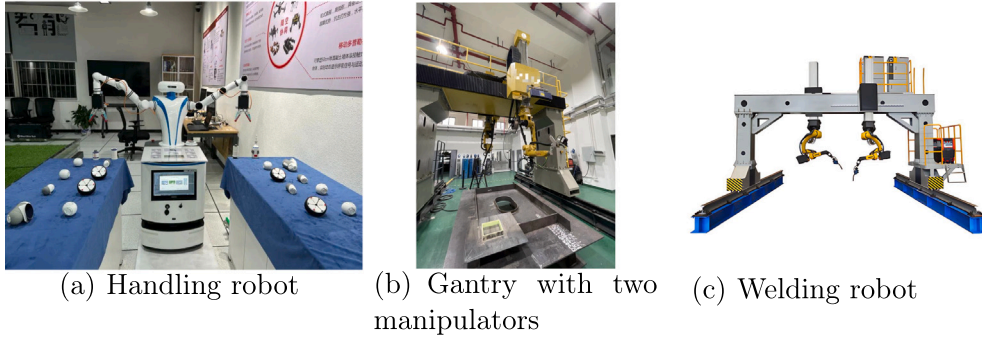


Fig. 1. Three cases with MRTALS problems.

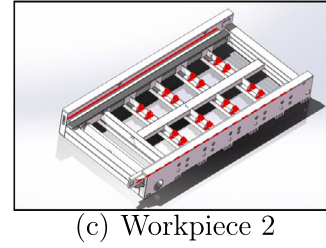
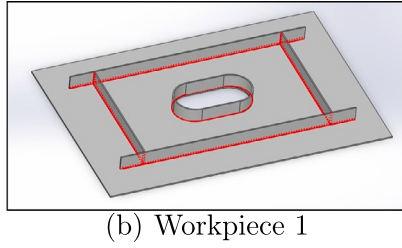
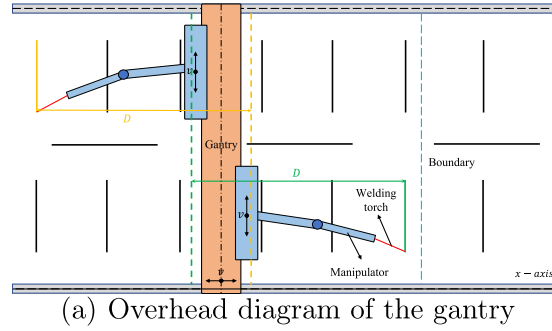


Fig. 2. An illustration of a gantry associated with two robots.

wherein the cost (i.e. cycle time) associated with a particular robot-task pair depends not only on the tasks assigned to the robot but also upon the schedules of other robots; (ii) complex dependencies, where the cost of a given robot-task pair is influenced by the schedules of other robots, such as those moveable gantries. In contrast to MRTA-related problems, wherein the multiple traveling salesman problem has been leveraged to optimize the sequencing of nodes (such as welding spot or drill) for each robot (Chakraa et al., 2023), the MRTALS necessitates the scheduling of robots for weld lines, where each line is conceptualized as an arc within the graph.

The MRTA has been widely used to formulate many practical applications, such as spot-welding, disassembly and arc-welding. One notices that many efforts have concentrated on the MRTA for spot-welding (Wang et al., 2023; Zhao et al., 2023; Zhou et al., 2022). Lopes et al. (2017) addressed the issue of balancing a robotic spot-welding production line, modeling it after a real-world car plant in Curitiba, Brazil, using a mixed-integer linear programming model. Similarly, Lee et al. (2022) developed a disassembly sequence planning algorithm for task allocation in disassembly processes, establishing a cost function based on total disassembly time and employing a planning method to minimize it. In addition, Yin et al. (2022) investigated a partial disassembly line balancing problem with multi-robot workstations, formulating a cost function that considers cycle time, energy consumption, and hazard index improvement. They proposed a multi-objective hybrid driving algorithm utilizing a three-layer coding heuristic method to minimize costs. To address heterogeneous robots' order fulfillment

systems, Wang et al. (2020) proposed an optimal method based on a mixed-integer linear programming model to manage complex scheduling constraints and the coupled temporal-spatial relationships between robots. Based on the construction and weighted matching of bipartite graphs, an online integer linear programming algorithm is proposed to solve the task assignment problem of multiple UAVs (Ghassemi & Chowdhury, 2022). Furthermore, Shorinwa et al. (2023) introduced a distributed algorithm for solving multi-robot task allocation problems, where each robot addresses local optimization problems and shares the results with neighboring robots. Different with above studies, the MRTA are seldom used to schedule multiple robots to carry out arc-welding. Lee et al. (2023) presented a real-world welding gantry robot scheduling problem at shipyards and a mixed-integer linear programming model was proposed to solve the problem. Kim et al. (2002) developed a genetic algorithm to determine welding task sequencing for arc-welding. Kim et al. (2005) proposed several heuristic algorithms to optimize arc-welding task sequencing sequences of robots while avoiding heat-caused distortion. Indeed, the maximum working span of manipulators has a significant impact on task allocation when scheduling robots in many real-world applications, and no studies have associated this feature to the MRTA. In addition, one notices that all of these studies focus on task allocation and other constraints, such as heat caused distortion, without considering working span of each robot. In this paper, we fill this gap by studying the MRTA with limited span and proposing an effective algorithm to solve it.

Notices that the MRTA is generally formulated as a $\mathcal{NP}$ -hard problem, many metaheuristic algorithms are proposed to solve it efficiently.

For instance, Zhou et al. (2022) proposed a genetic algorithm to segment large workpieces into smaller pieces, each of which can be treated as a traveling salesman problem and subsequently optimized using the Lin–Kernighan–Helsgaun solver (Helsgaun, 2000). Similarly, Martin et al. (2023) proposed a cooperative game theory approach, employing Shapley values to decompose MRTA problems into manageable subproblems. Liu et al. (2023) introduced a two-step task allocation method combined with a scheduling algorithm to effectively address workload sharing and precedence constraints. For the logistics-oriented MRTA, Contini and Farinelli (2021) proposed a decentralized coordination algorithm based on a token-passing method, facilitating conflict-free path generation for multi-robot systems tasked with deliveries. In order to solve the MRTA with energy consumption constraint, Djenadi et al. (2024) proposed a lexicographic optimization-based multi-robot task allocation (LO-MRTA) approach. In scenarios involving non-mandatory functional tasks, Yan and Di (2023) utilized particle swarm optimization to solve the problem. Additionally, Mutti et al. (2021) presented an iterative hierarchical optimization method that integrates whale optimization and ant colony optimization algorithms. For a comprehensive review related to solution approach, readers can refer (Chakraa et al., 2023).

One notices that existing algorithms are mainly based on swarm intelligence optimization, such as particle swarm optimization (Yan & Di, 2023), whale optimization and ant colony optimization (Mutti et al., 2021). Based on this observation, our work is motivated by the following. First, the MRTALS generalizes the MRTA and inherits the computational complexity and it is also classified as a $\mathcal{NP}$ -hard problem (Chakraa et al., 2023). Second, the MRTALS aims to determine the sequences of arcs can be considered as a variant of the arc routing problem, which poses a computational challenge for solving methods. Third, existing algorithms for the MRTA are mainly based on swarm intelligent algorithms, which are not suitable for solving the MRTALS. In this paper, we fill this gap by proposing an effective hybrid genetic algorithm (HGA) with several dedicated components to improve the solution approaches for the MRTALS. The algorithm has been widely used to solve several routing problems, such as the vehicle routing problem (He & Hao, 2023) and the arc routing problem (Arakaki & Usberti, 2018), with very excellent performances. In addition, we conduct extensive experimental investigations to evaluate the performance of HGA on real-world instances. The contributions of this paper can be summarized as follows:

- We introduce a new variant of multi-robot task allocation (MRTA), the MRTA with limited span (MRTALS), motivated by real-world applications where a robot's working span is limited by a movable gantry. Formulated as a mixed-integer programming model for precise representation, the framework enhances automation systems, supports diverse industrial applications, such as robotic material handling, flexible manufacturing systems and aerospace and shipbuilding industries, and lays the groundwork for future advances in smart and sustainable manufacturing.
- We propose an effective hybrid genetic algorithm (HGA) to solve the MRTALS. The algorithm first incorporates a dedicated region division method to divide workpieces into regions where manipulators can access all weld lines within each region, which can reduce the search space. Second, the algorithm uses a new route-based crossover to generate promising offspring solutions to improve diversification and prevent the algorithm from converging too quickly. Third, an effective neighborhood-based local search procedure is used to improve each offspring solution as much as possible. The crossover and local search achieve a balance the diversification and intensification, which can help the algorithm have a substantial ability to find the better results.
- Computational experiments conducted on a set of real-world instances illustrate the competitiveness of HGA. The results indicate that the MRTALS can significantly reduce the cycle time. Additionally, the results from HGA have been adapted for application in real-world assembly lines.

The remainder of this paper is organized as follows. Section 2 introduces the mathematical formulation for the MRTALS. Section 3 provides a detailed insight into the proposed algorithm. Section 4 presents the computational results on real-world instances. Finally, Section 5 concludes the paper and discusses future research perspectives.

2. Problem formulation

As shown in Fig. 1(b), a workpiece is transferred to the gantry and secured in place. The workpieces predominantly consist of plate units, as illustrated in Fig. 2(b–c). It is observed that when the gantry is stationary, two manipulators (robots) can weld lines, because the simultaneous movement of torches and gantry can significantly compromise robot control accuracy. Once the gantry stops, the torches can weld lines situated within their respective boundaries, where each manipulator is limited by a specific working span. Following the completion of welding, the gantry moves and pauses, allowing the manipulators to resume welding operations. This iterative process continues until all lines are welded by the torches. The objective of the MRTALS is to determine the welding sequences of robots, and the movement time and distance of the gantry, to minimize cycle time, while accounting for the limited working span of each robot. The cycle time encompasses welding time, travel time of torches, gantry movement time, and waiting time. Notably, waiting time refers to the idle time when robots await others before the gantry moves. Given a set of arcs (weld lines) in the graph, once the gantry pauses, the routes of manipulators can be formulated as a routing problem, with the objective of minimizing the maximum length among a set of routes. One notes that the minmax multiple traveling salesman problem (mTSP) cannot be utilized to formulate the MRTALS, because the minmax mTSP is designed to plan the sequence of cities (nodes) (He & Hao, 2023), whereas the MRTALS aims to determine the sequence of arcs.

Given a set of weld lines $\mathcal{L} = \{1, \dots, l\}$ and a start point o , the MRTALS is defined on a directed complete graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, where $\mathcal{V} = \{o\} \cup \{1, \dots, n\}$ and $\mathcal{A} = \{(i, j) | i \in \mathcal{V}, j \in \mathcal{V}\}$. Each weld line $l \in \mathcal{L}$ is associated with two nodes $i_l \in \mathcal{N}$ and $j_l \in \mathcal{N}$, where node set $\mathcal{N} = \{1, \dots, n\}$. Additionally, each weld line has a welding time l_w . Each arc in $\mathcal{A} \setminus \mathcal{L}$ is associated with a non-negative traveling time c_{ij} , which satisfies the triangle inequality ($c_{ij} + c_{jk} > c_{ik}$ for all $i, j, k \in \mathcal{V}$ and $i \neq j \neq k$). There are $|\mathcal{K}|$ homogeneous manipulators, each starting from the start point o . Given that torches can only work when the gantry is stationary, each workpiece can be divided into a set of areas ($\mathcal{W} = \{1, \dots, w\}$), where weld lines in each area can be reached by manipulators. Given the maximum working distance (D) of manipulators along x -axis (shown in Fig. 2(a)), each workpiece can be divided into h areas $\mathcal{W}$, where $h = \lceil H/(2 * D) \rceil$ and H is the length of workpiece along x -axis. Let $\mathcal{L}_w$ be the corresponding weld lines where $w \in \mathcal{W}$, as well as $\mathcal{V}_w$ and $\mathcal{A}_w$. The moving time of the gantry toward x -axis is denoted by t_w , where $w \in \mathcal{W}$. The objective of the MRTALS is to minimize the cycle time, subject to the condition that each weld line is visited exactly once, which is equivalent to each node $i \in \mathcal{N}$ being visited exactly once. The MRTALS is associated with the following assumptions: (i) the distance between two nodes is calculated by Euclid distance without considering the manipulators' configurations; (ii) the movement speed of gantry and manipulators are constant. (iii) the time cost of collision avoidance is negligible. There are three decision variables used in the mathematical model. The binary variable $x_{ij}^k = 1$ if arc $(i, j) \in \mathcal{A}$ is visited by manipulator $k \in \mathcal{K}$. If $i \in \mathcal{V} \setminus \{o\}$ is visited by manipulator k , the binary $y_{ik} = 1$, otherwise not. For each node $i \in \mathcal{N}$ and manipulator $k \in \mathcal{K}$, there is a time variable u_{ik} specifying the cumulative time at node i when visited by manipulator k . The arc-flow model based on 3-index for the MRTALS is formulated as follows:

$$\text{Min } f = \max(u_{ik}^v) \quad (1)$$

subject to:

$$y_{ik} = 1 \quad i \in \mathcal{N} \quad (2)$$

$$\sum_{i \in \mathcal{N}} x_{io}^k = \sum_{j \in \mathcal{N}} x_{oj}^k = 1 \quad k \in \mathcal{K} \quad (3)$$

$$\sum_{k \in \mathcal{K}} x_{ijl}^k = 1 \quad l \in \mathcal{L} \quad (4)$$

$$\sum_{i \in \mathcal{V} \setminus \{a\}} x_{ia}^k = \sum_{j \in \mathcal{V} \setminus \{a\}} x_{aj}^k = y_{ak} \quad a \in \mathcal{N} \quad (5)$$

$$u_{ik}^w - u_{jk}^w + Mx_{ij}^k \leq M - c_{ij} \quad i \in \mathcal{N}, j \in \mathcal{N}, w \in \mathcal{W} \quad (6)$$

$$u_{ik_1}^w = \max(u_{j,k_2}^{w-1}) + t_w \quad i \in \mathcal{N}_{w-1}, j \in \mathcal{N}_w, k_1 \in \mathcal{K}, k_2 \in \mathcal{K} \quad (7)$$

$$x_{ij}^k \in \{0, 1\} \quad i \in \mathcal{V}, j \in \mathcal{V}, k \in \mathcal{K} \quad (8)$$

$$y_i^k \in \{0, 1\} \quad i \in \mathcal{V}, k \in \mathcal{K} \quad (9)$$

$$u_{ik}^w \in \mathcal{R}_0^+ \quad i \in \mathcal{V}, j \in \mathcal{V}, k \in \mathcal{K} \quad (10)$$

The objective function (1) aims to minimize the maximum time taken for the last manipulator to finish its allocated weld lines. Constraints (2) ensure that each node must be visited exactly once by manipulators. Constraints (3) ensures that each manipulator starts from the start node and return to the start node upon completing all tasks. Constraints (4) ensure that each weld line should visited exactly. Constraints (5) are flow conservation constraints. Constraints (6) correspond to vehicle-specific MTZ-formulation (Miller et al., 1960) and time constraints are used to eliminate subtours, where M is a big number. Constraints (7) ensure the cumulative time of continues two areas. Specifically, the gantry will move toward the next area with time t_w only when all manipulators pause working. Therefore, the starting time of manipulators will be the same with each other at the beginning of each area $w \in \mathcal{W}$. Constraints (8)–(10) define the decision variables. One notices that there are several ways to build the mathematical model of the MRTALS. For example, the MTZ (Miller–Tucker–Zemlin) formulation is used to eliminate subtours in the model. Alternatively, the DFJ (Dantzig–Fulkerson–Johnson) formulation can also be used to eliminate subtours. In fact, the first way requires a time variable u_{ik} in the model and the number of constraints is polynomial. However, the variable can be removed when using the DFJ formulation shown in Constraints (11), where $r(S)$ is the minimum number of manipulators needed to serve set S . Unfortunately, the number of constraints (11) is exponential and some cuts should be proposed to reduce the solution space. In fact, there are other ways to formulate the problem, such as the set partitioning model. In this paper, in order to present the mathematical model of MRTALS intuitively and concisely, we chose a 3-index and used the MTZ method to eliminate subtours.

$$\sum_{i \in S} \sum_{j \notin S} x_{ij}^k \geq r(S), \forall S \in \mathcal{V} \setminus \{0\}, S \neq \emptyset, k \in \mathcal{K} \quad (11)$$

Given the inherent non-linearity of constraints (7), mathematical programming solvers, such as Cplex, cannot directly solve the mathematical model. Indeed, the reason for using this constraint in the model is that each manipulator has a limited working span, so a large workpiece should be divided into several areas. In each area, each task can be reached by one manipulator. Unlike the MRTA, which does not have the limited working span constraint, all tasks can be scheduled together to obtain a good schedule. In practical applications, this condition cannot be satisfied because manipulators usually have a limited span. It is noticed that there are no general mathematical models that includes this feature. Since the mathematical model can only solve small instances, and there are no effective algorithms to deal with large instances, we need to design a new algorithm to solve the MRTALS by effectively partitioning set $\mathcal{L}$ into w subsets ($\mathcal{L}_1$ to $\mathcal{L}_w$).

In fact, the subdivision of the workpieces is a critical aspect of the designed algorithm that requires careful consideration because a good subdivision can improve the performance of the algorithm. To address this challenge, we propose a highly effective hybrid genetic algorithm tailored specifically tailored to tackle the MRTALS. By exploiting the strengths of hybrid genetic algorithms, our approach provides a robust solution methodology for optimizing the MRTALS.

3. Solution approach

In this section, we introduce a highly effective hybrid genetic algorithm (HGA) designed to solve the MRTALS. As shown in Fig. 3, the algorithm, also outlined in Algorithm 1, comprises a dedicated region division algorithm (line 2) responsible for dividing each workpiece into discrete regions, and an *optimization* engine (lines 3–15) consisting of several interrelated components dedicated to solving the MRTALS. The region division algorithm employs heuristic methods to divide all weld lines ($\mathcal{L}$) into subsets $\mathcal{L}_w$. For each subset $\mathcal{L}_w$, an initial population $\mathcal{P}_w$ is constructed, with each initial solution generated through a population initialization procedure (line 4). Subsequently, each population undergoes evolution over multiple generations, facilitated by three search operators: crossover, local search, and a population updating procedure. Following the generation of each offspring solution via route-based crossover (line 8), the local search procedure (line 9) further refines it to the fullest extent. Finally, all solutions within the population are managed and updated using a pool updating method (line 13). For the optimization engine, it stops and returns the best solution φ_w^* corresponding to subset $\mathcal{L}_w$ as soon as a predefined stopping condition is met. The algorithm concludes when all weld lines are considered by the optimization engine. The systematic approach ensures efficient exploration of the solution space and robust convergence towards sub-optimal solutions for the MRTALS. The proposed algorithm demonstrates significant advantages in addressing the challenges of MRTALS. By employing a region division strategy, it effectively overcomes the complexities associated with limited span constraints. Furthermore, the combination of route-based crossover and local search achieves a well-balanced trade-off between diversification and intensification, ensuring both exploration of the solution space and refinement of high-quality solutions. This design enhances the algorithm's efficiency and robustness in solving complex multi-robot task allocation problems.

3.1. The region division algorithm for large-sized workpieces

One notices that if the length (H) of a workpiece along x -axis is greater than the maximum working distance ($2 * D$) of manipulators, weld lines belonging to the workpiece should be divided into at least $w = \lceil H/(2 * D) \rceil$ subsets, where weld lines belonging to a subset can be welded by manipulators without moving the gantry. Therefore, we propose a dedicated region division algorithm to heuristically divide all weld lines ($\mathcal{L}$) into subsets.

As the gantry can only move along the x -axis, the region division algorithm seeks to determine the appropriate position for the gantry. In other words, the algorithm aims to divide $\mathcal{L}$ into several subsets. For each weld line $l \in \mathcal{L}$, let $x_l = \max(x_{i_l}, x_{j_l})$ and $x'_l = \min(x_{i_l}, x_{j_l})$, where x_{i_l} and x_{j_l} represent the coordinates along the x -axis of nodes i_l and j_l , respectively. Let $\mathcal{T}$ be a list that stores the values of x_l for all weld lines, and the values in $\mathcal{T}$ are sorted in ascending order.

As illustrated in Algorithm 2, let D_c represent the leftmost position along the x -axis where manipulators need to reach. Let $D_c = \arg \min\{x'_l | x_l \in \mathcal{T}\}$. The algorithm operates as follows: Firstly, all weld lines are iterated over based on the values of $x_l \in \mathcal{T}$. If weld lines can be accessed by manipulators from the leftmost position D_c and within their working span ($2 * D$) along the x -axis, the corresponding x_l values are saved in $\mathcal{T}_a$ (lines 6–7). When weld lines cannot be covered by the

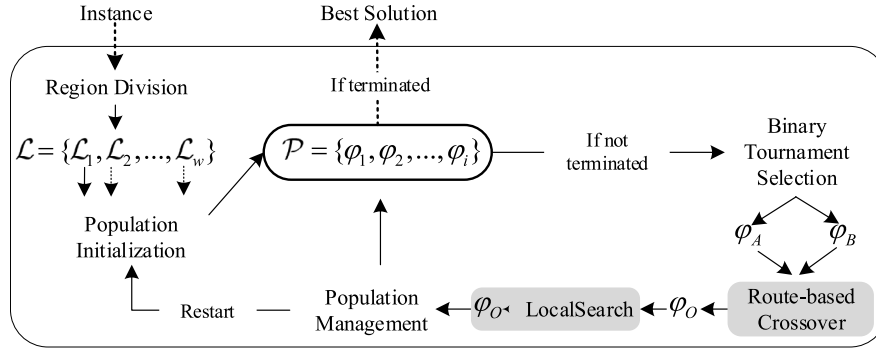


Fig. 3. Flow chart of the hybrid genetic algorithm.

Algorithm 1: Main framework of the HGA algorithm for the MRTALS

Input: Instance I ;
Output: The best solution φ^* found so far;

```

1 begin
2    $\mathcal{W} \leftarrow \text{RegionDivision}(I)$ ; /* Divide large-sized
   workpieces, Section 3.1 */
3   /* Optimization engine */
4   for  $w = 1$  to  $|\mathcal{W}|$  do
5      $P_w \leftarrow \text{PopulationInitial}(\mathcal{L}_w)$ ; /* Generate an
   initial population, Section 3.2.1 */
6      $\varphi_w^* \leftarrow \arg \min \{f(\varphi_i) | i = 1, 2, \dots, |P_w|\}$ ; /*  $\varphi_w^*$  Record
   the best solution found so far for  $\mathcal{L}_w$  */
7     while Stopping condition is not met do
8        $\{\varphi_A, \varphi_B\} \leftarrow \text{SelectParent}(P_w)$ ;
9        $\varphi_O \leftarrow \text{Crossover}(\varphi_A, \varphi_B)$ ; /* Generating
   offspring solutions, Section 3.2.2 */
10       $\varphi_O \leftarrow \text{LocalSearch}(\varphi_O)$ ; /* Improving the
   offspring solution, Section 3.2.3 */
11      if  $f(\varphi_O) > f(\varphi_w^*)$  then
12         $\varphi_w^* \leftarrow \varphi_O$ ;
13       $P_w \leftarrow \text{PoolUpdating}(P_w, \varphi_O)$ ; /* Managing the
   population, Section 3.2.4 */
14    end
15  end
16  return  $\varphi^* \leftarrow \{\varphi_1^*, \dots, \varphi_w^*\}$ ;
17 end

```

manipulators without moving the gantry toward x-axis (line 9), the loop breaks and stores the corresponding weld lines in $\mathcal{L}_w$ (line 25).

Given that manipulators are typically installed on both sides of the gantry and the objective of the MRTALS is to minimize the maximum cycle time, the gantry should be positioned centrally among all covered weld lines in each region to reduce the manipulators' waiting time as much as possible. As shown in lines 11 to 23 in Algorithm 2, we propose a heuristic strategy to balance the workload on both sides of the gantry when dividing weld lines into regions. For a given threshold τ , if the weld lines can be reached between $D_c + 2D\tau$ and $D_c + 2D$, we attempt to find the best configuration for these weld lines. Generally, τ is set to 0.8. A smaller τ results in dividing each workpiece into too many regions, while a larger τ makes it difficult to ensure balanced workloads between the two sides of the gantry. During this process, g_l and g_r are used to record the welding time of weld lines accessible to manipulators on the left and right sides of the gantry, respectively. Let l_i be the welding time of weld line i . If the absolute difference of

Algorithm 2: Region division Algorithm

Input: List $\mathcal{T}$, Parameters τ and ξ ;
Output: $\{\mathcal{L}_1, \dots, \mathcal{L}_w\}$;

```

1 begin
2    $w \leftarrow 0$ ;
3   while  $\mathcal{T} \neq \emptyset$  do
4      $\mathcal{T}_a \leftarrow \emptyset$ ;  $D_c = \arg \min \{x_i | x_i \in \mathcal{T}\}$ ;
5     for  $l = 1$  to  $|\mathcal{T}|$  do
6       if  $x_l \leq 2 * D + D_c$  then
7          $\mathcal{T}_a \leftarrow \mathcal{T}_a \cup \{x_l\}$ ;
8       else
9         Break;
10      end
11      if  $D_c + 2D\tau \leq D_c + x_l \leq 2D$  then
12         $g_l \leftarrow 0$ ;  $g_r \leftarrow 0$ ;
13        for  $i = 1$  to  $|\mathcal{T}_a|$  do
14          if  $x_i - D_c \leq (x_l - D_c)/2$  then
15             $g_l \leftarrow g_l + l_i$ ; /* Computing the welding
            time of lines at the left side of
            the gantry; */
16          else
17             $g_r \leftarrow g_r + l_i$ ; /* Computing the welding
            time of lines at the left side of
            the gantry; */
18          end
19        end
20        if  $|g_l - g_r| / \max(g_l, g_r) \leq \xi$  then
21          break;
22        end
23      end
24    end
25     $\mathcal{L}_w \leftarrow \{l | x_l \in \mathcal{T}_a\}$ ;  $w \leftarrow w + 1$ ;  $\mathcal{T} \leftarrow \mathcal{T} \setminus \mathcal{T}_a$ ;
26  end
27  return  $\{\mathcal{L}_1, \dots, \mathcal{L}_w\}$ ;
28 end

```

welding time visited by manipulators on both sides is less than ξ , the weld lines in $\mathcal{T}_a$ can be divided into a new region. In this paper, we set $\xi = 0.1$. As shown in Fig. 4(a), we can easily find that the welding time of red (g_l) and blue (g_r) lines are roughly equal when $x_l = 2D * \tau$. However, the welding time of blue lines is greater than red lines too much. Therefore, the first region only consists of all weld lines where $x_l \leq 2D * \tau$. After dividing the first subset $\mathcal{L}_1$, all weld lines in $\mathcal{T}$ are removed, as shown in Fig. 4, and the algorithm continues to divide the remaining weld lines.

3.2. Optimization engine

After dividing set $\mathcal{L}$, $|\mathcal{W}|$ subsets $(\mathcal{L}_1, \dots, \mathcal{L}_w)$ are obtained. For each subset $\mathcal{L}_w$, the optimization engine is employed to determine the best

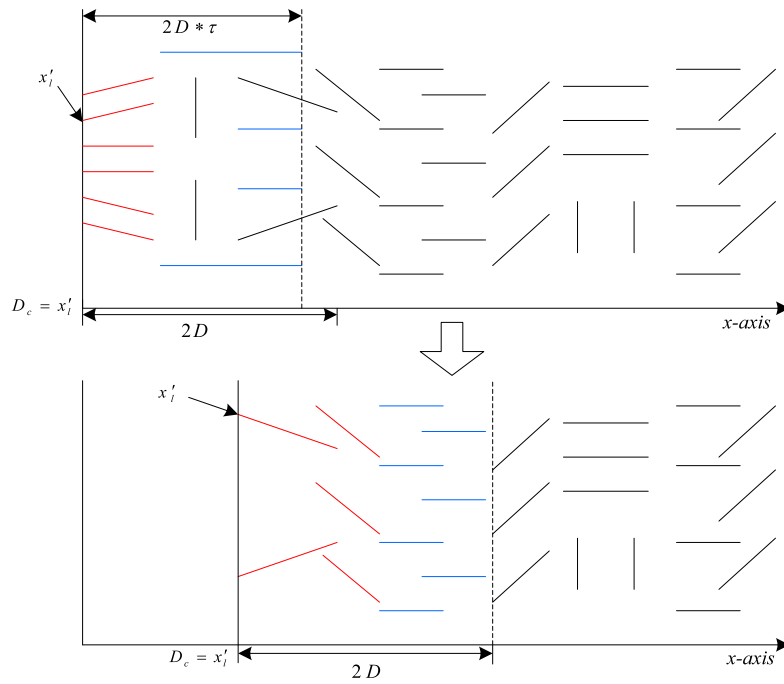


Fig. 4. An example of using region-division algorithm to divide weld lines.

task allocation for each manipulator. Within the *optimization* engine, the population is initially constructed, and subsequently evolved through the application of crossover, local search, and population management techniques. It is important to note that for each subset $\mathcal{L}_w$, the welding order for each manipulator can be conceptualized as a single tour.

3.2.1. Population initialization

The HGA algorithm initiates its search with a population $\mathcal{P}_w$ comprising μ initial solutions. The construction process of each solution consists of three steps. Firstly, $|\mathcal{K}|$ tours are initialized, each starting from the designated start node o . Secondly, a random unassigned weld line $l \in \mathcal{L}_w$ is selected and inserted into the shortest route at the position resulting in the least increase in route length. It is important to note that if the weld line is undirected, offering two possible insertion directions into the route. Once all weld lines in $\mathcal{L}_w$ are assigned, a feasible solution is obtained. Finally, the solution undergoes refinement via the local search procedure (Section 3.2.3) before being added to the corresponding population $\mathcal{P}_w$. The initialization procedure stops once μ solutions have been obtained.

3.2.2. Route-based crossover

At each generation of HGA, a crossover operator is used to generate a new offspring solution by combining two parent solutions selected from the population. It is widely acknowledged that the efficacy of a population-based hybrid genetic algorithm critically hinges on the crossover operator. The operator should not only produce diverse solutions but also effectively inherit meaningful building blocks from parents to offspring. In this paper, a dedicated route-based crossover is utilized to generate promising offspring solutions. Prior to applying the crossover operator, two parent solutions are selected from the population using a binary tournament strategy based on their objective values.

The route-based crossover operator, widely employed in solving various routing problems (He et al., 2021; Lu et al., 2019), appears well-suited for the MRTALS, given that a solution to the MRTALS comprises a set of routes. Given two parent solutions $\varphi_A = \{r_A^1, r_A^2, \dots, r_A^k\}$ and $\varphi_B = \{r_B^1, r_B^2, \dots, r_B^k\}$, the route-based crossover essentially involves copying φ_A to create an offspring solution φ_O , and then replacing a route of φ_O with a route from φ_B . Subsequently, φ_O is repaired to

ensure feasibility. The crossover procedure encompasses the following steps:

- Step 1: All routes from φ_A are copied to create an offspring solution φ_O , and then route r_O^a is selected from φ_O and replaced with route r_B^b selected from solution φ_B , where a and b are randomly selected integers between 1 and $|\mathcal{K}|$. Subsequently, all weld lines that are not in φ_O are collected into a list S .
- Step 2: All duplicated weld lines in the remaining routes $r \in \varphi_O \setminus r_O^a$ are removed.
- Step 3: The weld lines from S are inserted into φ_O . Before insertion, the list is randomly shuffled. Each weld line in S is inserted it at the optimal position within the remaining routes with respect to the minimizing cycle time.

The route-based crossover operator not only introduces a new route (obtained from φ_B) into an offspring solution φ_O , but also modifies other existing routes due to the deletion of duplicated weld lines and the insertion of weld lines (that are not in φ_O). The crossover could lead to an offspring solution that is structurally different from its parent solutions.

The route-based crossover operator not only introduces a new route (derived from φ_B) into an offspring solution φ_O , but also modifies existing routes by removing duplicated weld lines and inserting weld lines that are absent in φ_O . As a result, the crossover operator can yield an offspring solution that diverges structurally from its parent solutions, which enhances the algorithm's capacity to explore diverse solution spaces and facilitates the discovery of novel and potentially superior solutions.

3.2.3. Local search

Local search stands as a core component within state-of-the-art heuristic algorithms designed to tackle various routing problems (Ren et al., 2022). Efforts to enhance the efficiency of local search and generate high-quality solutions have led to the development of advanced neighborhood operators and speed-up techniques. In this paper, our HGA adopts six commonly used neighborhood operators, as described in He and Hao (2023), and explores them within the framework of variable neighborhood descent.

To provide an intuitive understanding of these neighborhood operators, we introduce the following notations: $r(u)$ and $r(v)$ denote the routes that visit weld lines u and v , respectively. Let v be a neighbor of u , and let x and y represent the successors of u in $r(u)$ and v in $r(v)$, respectively. The substring from weld line u to x is denoted as (u, x) , and the substring from weld line v to y is denoted as (v, y) . Additionally, let r_m denote the longest route.

- M1: If $r(u) \neq r(v)$ and $r(u)$ is the longest route r_m , then remove u and place it after weld line v .
- M2: If $r(u) \neq r(v)$ and one of them is the longest route r_m , then swap u and v .
- M3: If $r(u) \neq r(v)$ and one of them is the longest route r_m , then swap two consecutive weld lines (u, x) and (v, y) .
- M4: The 2-opt operator, which replaces (u, x) and (v, y) by (u, v) and (x, y) if $r(u) = r(v)$.
- M5: The 2-opt* operator, which replaces (u, x) and (v, y) by (u, v) and (x, y) if $r(u) \neq r(v)$.
- M6: The 2-opt* operator, which replaces (u, x) and (v, y) by (u, y) and (v, x) if $r(u) \neq r(v)$.

It is noteworthy that M1 corresponds to the *Relocate* operator, while M2 and M3 correspond to the *Swap* operator. M4 represents the 2-opt operator, while M5 and M6 denote the 2-opt* operator. Furthermore, to streamline the search process and avoid examining too many non-promising candidate solutions, we employ the α -nearness technique. Specifically, for a weld line u , we only consider its α neighbor weld lines. This approach helps focus the search on the most relevant and potentially fruitful regions of the solution space.

3.2.4. Population management

In the context of utilizing HGA for solving various routing problems, an advanced survival strategy is indispensable (He & Hao, 2023). In this paper, we adopt a state-of-the-art technique for updating the population $\mathcal{P}$. Following a similar approach as He and Hao (2023), the algorithm continuously augments the population by incorporating offspring solutions until the size of $\mathcal{P}_w$ reaches $\mu + \lambda$, where λ denotes the generalized size parameter. Subsequently, the survival strategy is invoked to remove λ solutions based on their contributions to both the diversification of the population and the objective. Similar to He and Hao (2023), the normalized Hamming distance between a pair of solutions, denoted as φ_A and φ_B , is defined as the ratio of the number of non-common edges to the total number of edges in φ_A and φ_B . Subsequently, the biased fitness of each solution is computed, taking into account its fitness value and diversity rank within $\mathcal{P}_w$. This approach ensures a balanced selection of solutions that not only excel in terms of objective attainment but also contribute to population diversity, thus enhancing the algorithm's exploration–exploitation trade-off.

4. Computational results and comparisons

In this section, we evaluate the performance of HGA when solving three different sized instances: Inst1, Inst2 and Inst3.

- Inst1: The instance includes 54 weld lines. The case is an artificial instance, where the coordinate information of the end points of the lines is shown in Table A.1.
- Inst2: The instance includes 24 weld lines, as shown in Fig. 11, where the instance comes from Shanghai Friendess Electronic Technology Company, Ltd. The coordinate information of weld lines is shown in Table A.2.
- Inst3: The instance includes 132 weld lines. Same with Inst2, the instance also comes from the company and the coordinate information of the end points of the lines is shown in Table A.3.

4.1. Experimental protocol and reference methods

Parameters setting. HGA has three parameters: population size μ , generation size λ , and nearest neighbor granularity α . To calibrate the three parameters, we performed a sensitivity analysis. We identified a rough value range of values for each parameter and then analyzed one parameter at a time. Given that the size of instances is similar to He and Hao (2023), we set $\mu = 20$, $\lambda = 10$, $\alpha = 20$ as their default values. Specifically, we varied the values of each parameter within its range while keeping the other parameters at their default values, as shown in Table 1. All experiments were based on the three different sized instances introduced above. For each instance, we ran HGA 20 times with each parameter setting, and the results are shown in Fig. 5, where Avg_{gap} is defined in Eq. (12).

$$Avg_{gap} = \frac{F_{PS_{avg}} - F_{HGA_{avg}}}{F_{HGA_{avg}}} \quad (12)$$

The results in Fig. 5 illustrate the effect of the three parameters on the computational results. It can be observed that when the population size μ is set to 20, the algorithm's average performance of the algorithm Avg_{gap} approaches the best result. In addition, when the augmented population size λ is around 10, the result reaches its lowest point. As for nearest neighbor granularity, as long as α exceeds 15, its influence on the results becomes negligible. However, large values of α also consume computing time without contributing to better results. In summary, we conclude the values of three parameters and using them as default in the following experiments. In addition, the physical parameters of the gantry and manipulators are shown in Table 2.

Experimental setting and stopping criterion. The HGA algorithm is programmed in C++ and all experiments were conducted on a computer with 12th Gen Intel(R) Core(TM) i5-12490F processor of 3.0 GHz CPU and 16 GB RAM. Given the stochastic nature of the compared algorithms, HGA was run 30 times on each instance with different random seeds. The algorithm stops when it reaches a maximum of 200 000 iterations. One notices that each iteration means a solution is improved by the local search procedure.

Reference methods. Since the MRTALS is proposed in this paper, there are no widely applicable reference methods. In this study, we used two comparative methods to evaluate the effectiveness of the proposed HGA algorithm.

The first method is the expert-tuned method currently used by the company. The expert-tuned method used for this comparison comes from Shanghai Friendess Electronic Technology Company. The method integrates a greedy algorithm to generate an initial solution, which is then adjusted and improved based on the expert's domain knowledge. The method relies on human expertise to address special cases or scenarios not easily captured by automated methods, making it a practical benchmark.

The second method is called WAG+VNS, which was proposed by Lee et al. (2023) when solving a real-world welding gantry robot scheduling problem. Because of its relevance to multi-robot task allocation and scheduling, it serves as a representative approach for comparison in this paper. Since the source code of the algorithm is not available online, we implemented it based on the details provided in Lee et al. (2023) and made the necessary adaptations to solve the MRTALS problem. The algorithm was evaluated using the default parameters specified in Lee et al. (2023) and executed 30 times to select the best performing results for comparison with HGA.

These two comparative methods provide different benchmarks: the expert-tuned method emphasizes domain expertise and manual refinement, while WAG+VNS highlights the performance of an existing automated approach under the MRTALS scenario.

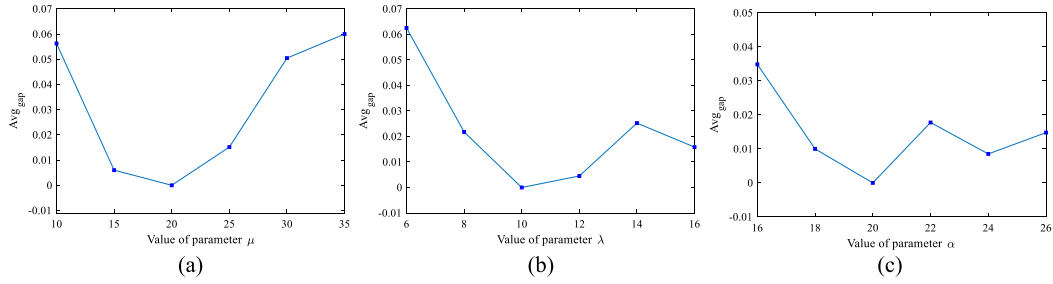


Fig. 5. Analysis of the effects of the parameters.

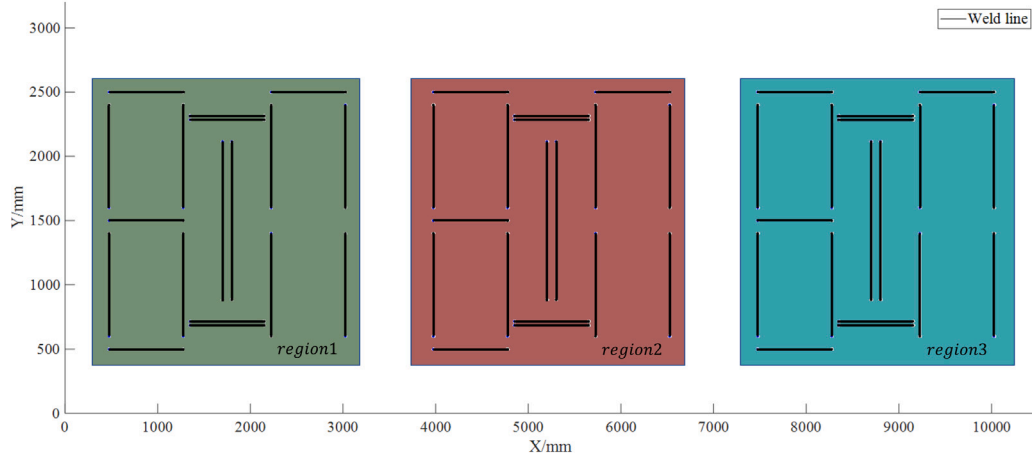


Fig. 6. The diagram of the region distribution result of instance Inst1.

Table 1
Settings of algorithm parameters.

Parameters	Section	Description	Candidate values	Values
μ	3.2.1	Size of the initial population	{10,15,20,25,30,35}	20
λ	3.2.4	Generalized augment size parameter	{6,8,10,12,14,16}	10
α	3.2.3	Number of weld lines in local search	{16,18,20,22,24,26}	20

Table 2
Physical parameters of the gantry and manipulators.

Parameters	Values	
	Inst1 & Inst2	Inst3
Welding speed of torches	30 mm/s	30 mm/s
Moving speed of torches	50 mm/s	50 mm/s
Moving speed of the gantry	50 mm/s	50 mm/s
The maximum working distance of manipulators (D)	1690 mm	1400 mm

Table 3
The cycle time of Inst1 obtained by three methods.

Methods	Cycle time (s)	Improve (%)
Expert-tuned method	1087.281	6.5
WAG+VNS	1041.080	2.4
HGA	1016.163	–

4.2. Inst1: An artificial instance

To evaluate the performance of HGA, we conducted experiments on Inst1, which consists of 54 weld lines (Fig. 6). Notably, all the weld lines in the instance can be naturally divided into three distinct regions. As shown in Fig. 6, three methods achieve the same results when dividing large workpiece. In addition, two manipulator paths generated by three methods are shown in Figs. 7–9, respectively. Detailed results of manipulators' routes for weld lines are shown in Tables B.1–B.3 in Appendix. As shown in Fig. 10 and Table 3, HGA outperforms the expert-tuned method and WAG+VNS, by reducing the cycle time by 6.5% and 2.4%, respectively. One notices that manipulator k_2 experiences significant waiting times when using the expert-tuned method, and on the contrary, this case is significantly minimized by HGA. In summary, HGA shows a competitive performance compared to the expert-tuned method and WAG+VNS.

4.3. Inst2: A simple cabin partition case test

In this section, we apply HGA to solve a small-sized real-world workpiece, referred to as Inst2, provided by Shanghai Friendless Electronic Technology Company, Ltd. As illustrated in Fig. 11, the workpiece is a bulkhead segment selected from a ship's cabin, featuring a total of 24 weld lines. In the factory, during the actual production process, workers first secure the individual components onto the base plate via spot welding. Subsequently, laser scanning and feature extraction techniques are employed to obtain the CAD model and coordinate information for all weld lines on the workpiece.

It is observed that the workpiece includes only 24 weld lines. When the gantry is positioned on the workpiece's central line along the x -axis, the maximum working distance of the manipulators can cover all weld lines, as the maximum length of the workpiece is 2500 mm, which is less than $2 * D$. Figs. 12–14 illustrate the welding sequences of the manipulators, respectively. As shown in Fig. 15 and Table 4, the cycle time achieved by the HGA is 1.8% less than that of the expert-tuned

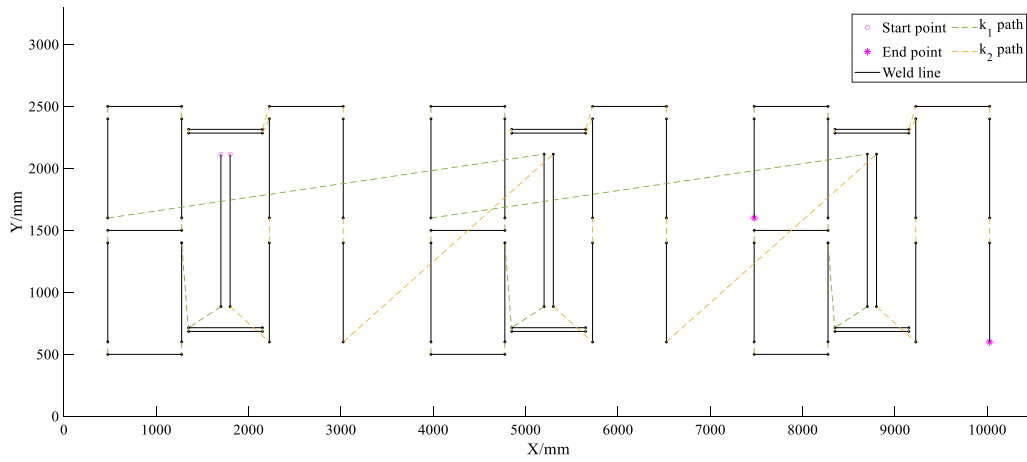


Fig. 7. The manipulators' path of Inst1 given by the expert-tuned method.

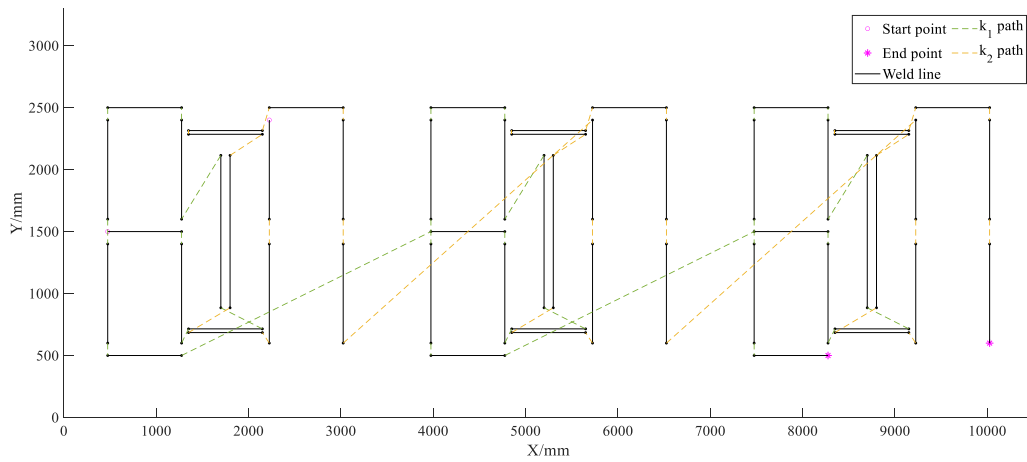


Fig. 8. The manipulators' path of Inst1 obtained by HGA.

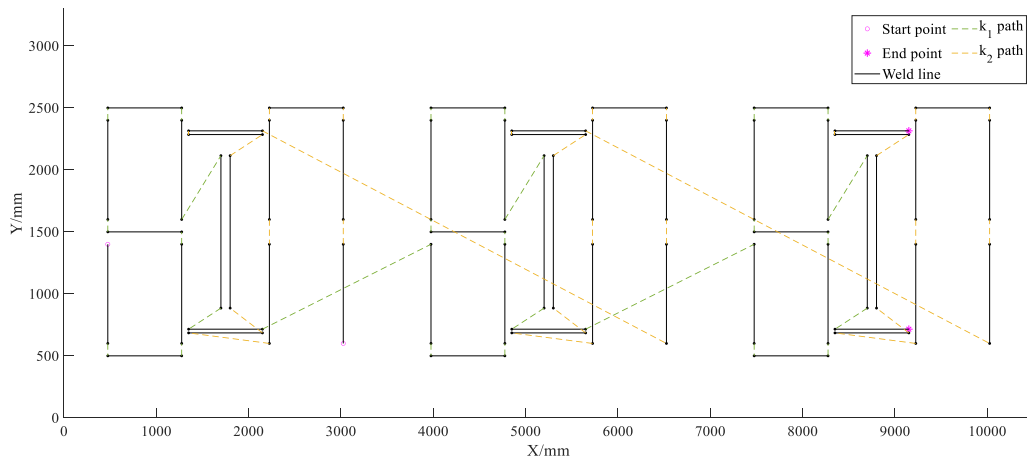


Fig. 9. The manipulators' path of Inst1 calculated by WAG+VNS.

method, and is 9.1% less than that of WAG+VNS. In addition, these improved results have been implemented on assembly lines to improve working efficiency.

4.4. Inst3: A complicated cabin partition case test

To further evaluate the performance of HGA and verify its adaptability to various scenarios, we applied the algorithm to solve a large

actual workpiece. As shown in Fig. 16, the workpiece is a complex structural component obtained from a ship's hull, featuring a total of 132 weld lines, including 130 short lines and 2 long lines that exceed the maximum working distance of the manipulators. Notably, the long weld lines are typically performed as the final welding process due to technical requirements where the gantry and manipulators move together. Although the area of this workpiece is not large, its internal structure is complex and the space is narrow. Consequently, a smaller

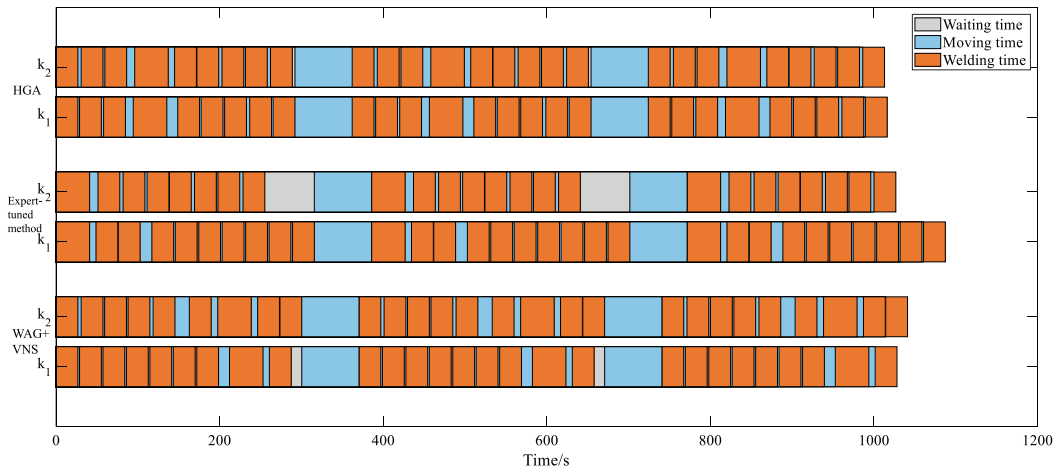


Fig. 10. The Gantt chart of manipulators' working time of Inst1.

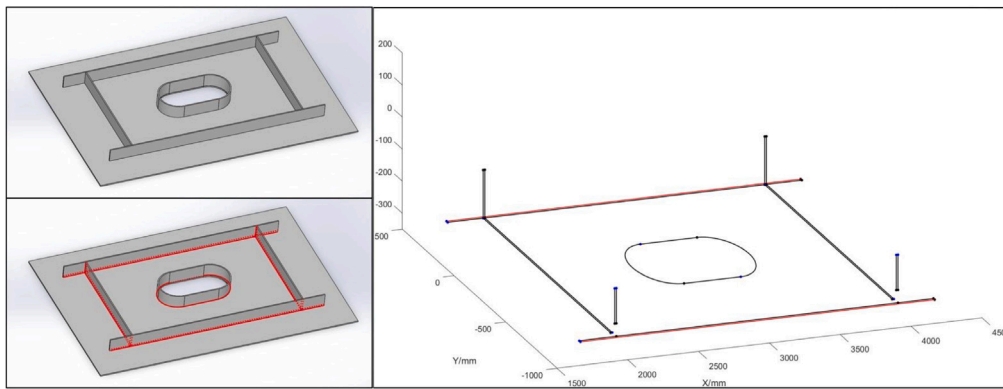


Fig. 11. The diagram of the simple plate unit workpiece.

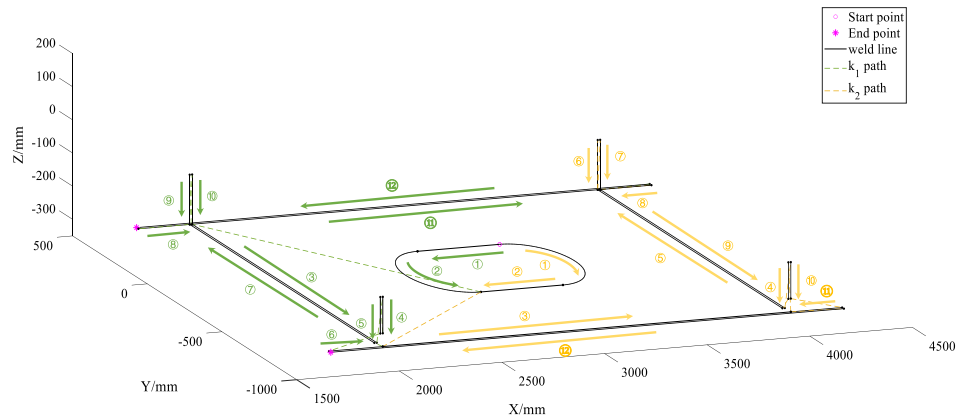


Fig. 12. The manipulators' path of Inst2 obtained by the expert-tuned method.

gantry and manipulators are used, with specific parameters listed in Table 2.

When assigning weld lines to manipulators, two long weld lines are assigned to a separate region. For the short weld lines, HGA and WAG+VNS divide them into two regions, as shown in Figs. 18 and 19. In contrast, the expert-tuned method divides the short welds into four regions (Fig. 17) and uses a symmetrical welding approach. For two long weld lines, it is worth noting that high-precision control during the welding process is theoretically possible because there are only

two long weld lines in this workpiece, which they are symmetrically distributed.

The final results are shown in Figs. 17–19, with the Gantt chart presented in Fig. 20. Under the same conditions, it is evident that the cycle time obtained from HGA is reduced by 9.9% compared to the expert-tuned method, and by 2.3% compared to WAG+VNS as shown in Table 5. Moreover, these improved results have been implemented on assembly lines, thereby enhancing overall working efficiency.

As shown in Tables 3–5, HGA demonstrates shorter cycle times and higher efficiency compared to the expert-tuned method. In addition, the

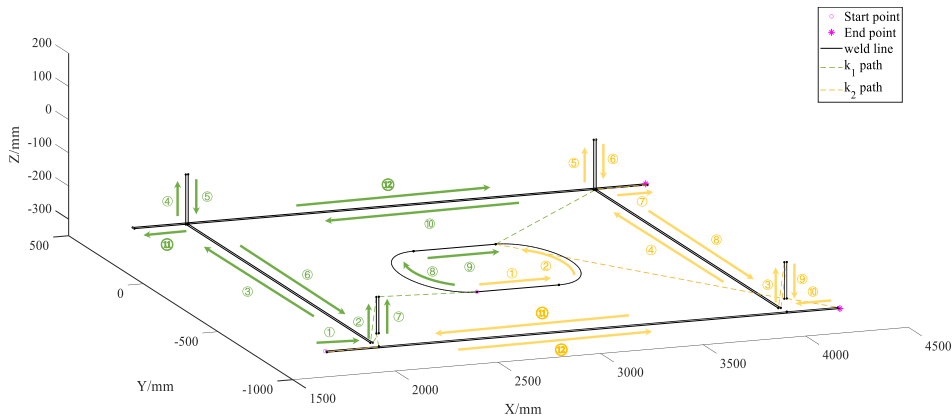


Fig. 13. The manipulators' path of Inst2 calculated by HGA.

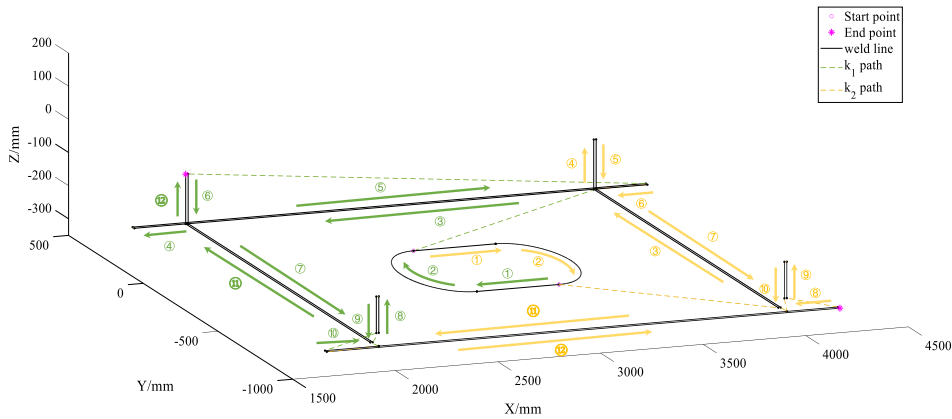


Fig. 14. The manipulators' path of Inst2 calculated by WAG+VNS.

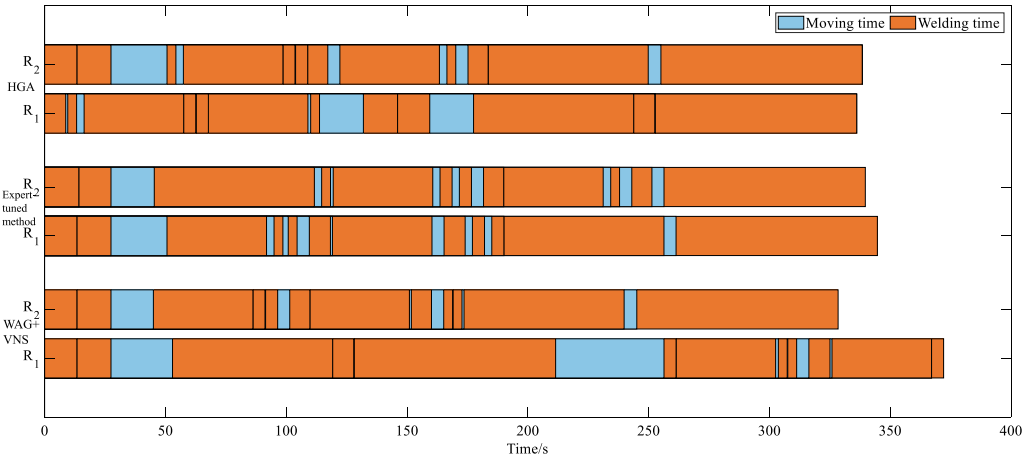


Fig. 15. The Gantt chart of manipulators' working time of Inst2.

Table 4
The cycle time of Inst2 obtained by three methods.

Methods	Cycle time (s)	Improve (%)
Expert-tuned method	344.690	1.8
WAG+VNS	372.104	9.1
HGA	338.421	–

Table 5
The cycle time of Inst3 obtained by three methods.

Methods	Cycle time (s)	Improve (%)
Expert-tuned method	731.929	9.9
WAG+VNS	674.726	2.3
HGA	659.465	–

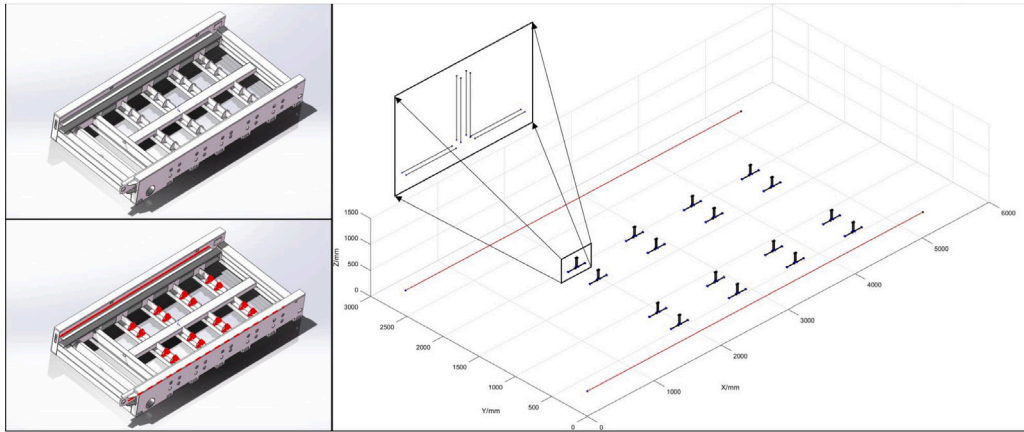


Fig. 16. The diagram of the complex structured workpiece.

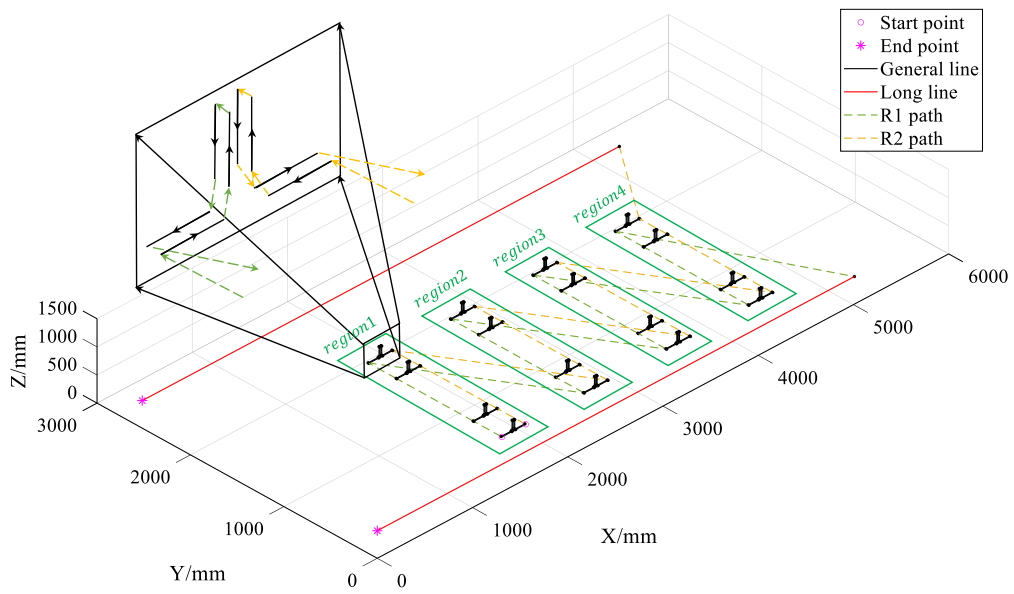


Fig. 17. The manipulators' path of Inst3 calculated by the expert-tuned method.

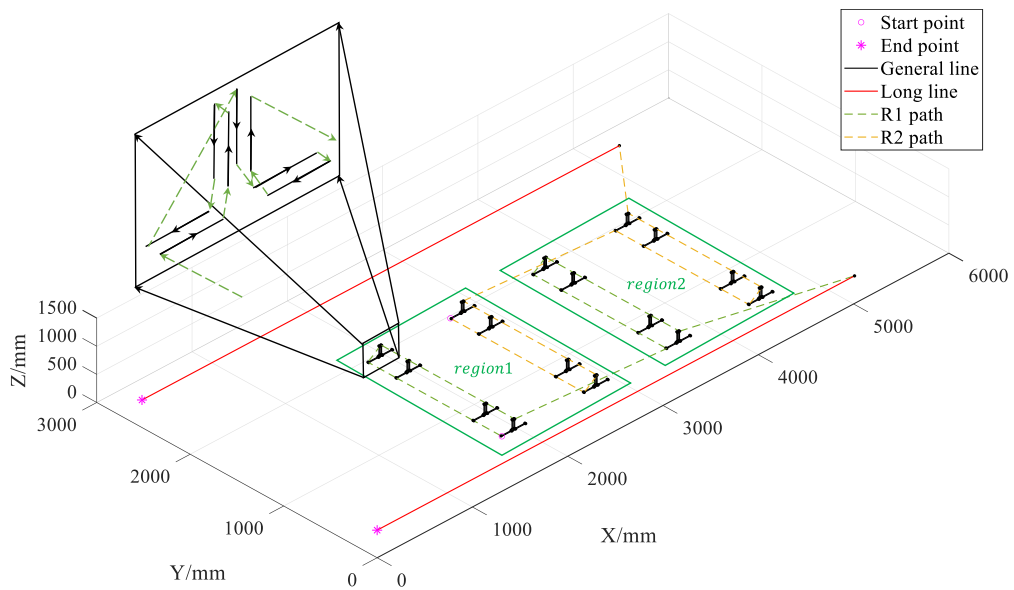


Fig. 18. The manipulators' path of Inst3 calculated by HGA.

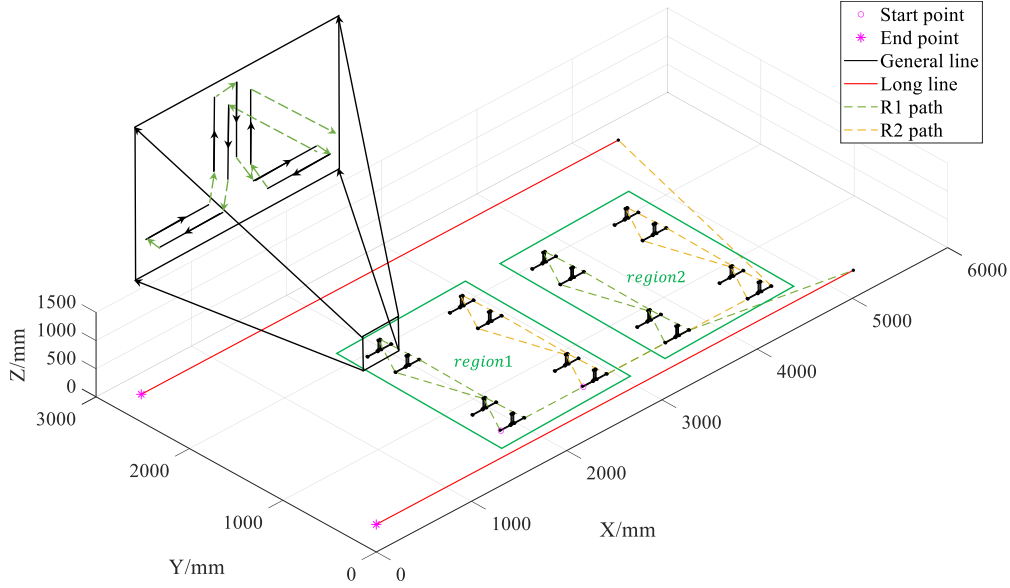


Fig. 19. The manipulators' path of Inst3 calculated by WAG+VNS.

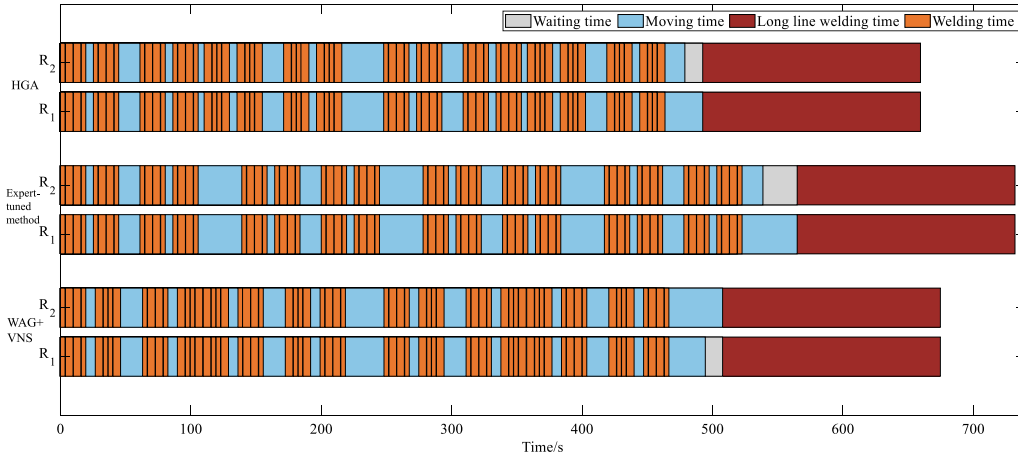


Fig. 20. The Gantt chart of manipulators' working time of Inst3.

Wilcoxon signed-rank test with confidence level of 0.05 is used to reveal the statistically significant difference between HGA and two other methods. Compare to the expert-tuned method, HGA achieves average improvement by 6.07% among three instances, and $p > 0.05$. Although there are no statistical significances between HGA and two reference methods, we can still summarize that HGA achieves much better results than two compared methods. For WAG+VNS, our algorithm improves 4.6% and $p > 0.05$. In addition, the results of both Inst2 and Inst3 have been implemented on assembly lines at Shanghai Friendess Electronic Technology Company, Ltd., to improve production efficiency.

5. Conclusion

In this paper, we addressed a new variant of the multi-robot task allocation problem: the multi-robot task allocation problem with limited span (MRTALS). The problem is derived from a real-world assembly line scenario where a set of industrial robots are installed on a movable gantry, each with a maximum working span. The objective is to minimize the cycle time required to complete a given set of welding lines. We thus present a mathematical model that captures this objective. Given the $\mathcal{NP}$ -hard of the problem, we propose an effective hybrid genetic algorithm (HGA) to solve it. The algorithm

incorporates a dedicated region division algorithm to divide large-sized workpieces into regions where all weld lines can be completed by manipulators without moving the gantry. Furthermore, a dedicated route-based crossover is used to generate diverse offspring solutions, and an effective neighborhood-local search procedure is employed to optimize each solution as much as possible.

Extensive computational experiments on three representative instances demonstrate the superiority of the proposed method, consistently outperforming the expert-tuned method and WAG+VNS in real-world assembly lines with significant efficiency improvements. The MRTALS problem, which models problems with constrained working spans for robots, is widely applicable across industries such as robotic material handling, flexible manufacturing systems, and aerospace and shipbuilding industries. The proposed method and framework not only optimize multi-robot task allocation but also hold the potential to drive innovation, enhance automation, and lay a foundation for intelligent and sustainable industrial systems.

While the proposed HGA demonstrates strong performance in optimizing MRTALS, the iterative nature of genetic algorithms can result in longer computation times, particularly for very large problem instances. Subsequently, future work can explore several directions. First, alternative algorithmic frameworks, such as tabu search or adaptive large neighborhood search, can be developed to address these

Table A.1

The information of all weld lines in instance Inst1.

Order	Start point			End point			Order	Start point			End point		
	X/mm	Y/mm	Z/mm	X/mm	Y/mm	Z/mm		X/mm	Y/mm	Z/mm	X/mm	Y/mm	Z/mm
1	475	2500	0	1275	2500	0	28	5725	2500	0	6525	2500	0
2	1275	1600	0	1275	2400	0	29	5725	1400	0	5725	600	0
3	475	1500	0	1275	1500	0	30	6525	600	0	6525	1400	0
4	475	600	0	475	1400	0	31	4850	2315	0	5650	2315	0
5	475	500	0	1275	500	0	32	4850	715	0	5650	715	0
6	1275	600	0	1275	1400	0	33	4850	2285	0	5650	2285	0
7	475	1600	0	475	2400	0	34	4850	685	0	5650	685	0
8	2225	1600	0	2225	2400	0	35	5200	2115	0	5200	885	0
9	3025	2400	0	3025	1600	0	36	5300	2115	0	5300	885	0
10	2225	2500	0	3025	2500	0	37	7475	2500	0	8275	2500	0
11	2225	1400	0	2225	600	0	38	8275	1600	0	8275	2400	0
12	3025	600	0	3025	1400	0	39	7475	1500	0	8275	1500	0
13	1350	2315	0	2150	2315	0	40	7475	600	0	7475	1400	0
14	1350	715	0	2150	715	0	41	7475	500	0	8275	500	0
15	1350	2285	0	2150	2285	0	42	8275	600	0	8275	1400	0
16	1350	685	0	2150	685	0	43	7475	1600	0	7475	2400	0
17	1700	2115	0	1700	885	0	44	9225	1600	0	9225	2400	0
18	1800	2115	0	1800	885	0	45	10025	2400	0	10025	1600	0
19	3975	2500	0	4775	2500	0	46	9225	2500	0	10025	2500	0
20	4775	1600	0	4775	2400	0	47	9225	1400	0	9225	600	0
21	3975	1500	0	4775	1500	0	48	10025	600	0	10025	1400	0
22	3975	600	0	3975	1400	0	49	8350	2315	0	9150	2315	0
23	3975	500	0	4775	500	0	50	8350	715	0	9150	715	0
24	4775	600	0	4775	1400	0	51	8350	2285	0	9150	2285	0
25	3975	1600	0	3975	2400	0	52	8350	685	0	9150	685	0
26	5725	1600	0	5725	2400	0	53	8700	2115	0	8700	885	0
27	6525	2400	0	6525	1600	0	54	8800	2115	0	8800	885	0

challenges, along with the creation of more comprehensive benchmark instances. Additionally, exact algorithms could be investigated to provide optimal solutions for specific cases. Second, the MRTALS framework can be extended to incorporate practical constraints, such as synchronized welding and heat impact considerations, enhancing its real-world applicability. Finally, broader societal applications of MRTALS, including improving manufacturing efficiency, reducing production costs, and advancing fields like aerospace, automotive, and shipbuilding, highlight its potential for significant industrial impact.

CRediT authorship contribution statement

Wenbo Liu: Writing – original draft, Visualization, Validation. **Zhian Kuang:** Project administration. **Yongcong Zhang:** Formal analysis. **Bo Zhou:** Investigation. **Pengfei He:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Shihua Li:** Data curation, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The information of three instances

See [Tables A.1–A.3](#).

Table A.2

The information of all weld lines in instance Inst2.

Case 2 ID	Start point			End point		
	X/mm	Y/mm	Z/mm	X/mm	Y/mm	Z/mm
1	1750	−872.933	−313.714	2012	−872.933	−313.714
2	2012	−872.933	−313.714	4000	−872.933	−313.714
3	4000	−872.933	−313.714	4250	−872.933	−313.714
4	2012	−832.933	−313.714	2012	403.066	−313.714
5	1750	403.066	−313.714	2012	403.066	−313.714
6	2012	403.066	−313.714	4000	403.066	−313.714
7	4000	403.066	−313.714	4250	403.066	−313.714
8	3988	403.066	−313.714	3988	−832.933	−313.714
9	1750	−884.933	−313.714	4250	−884.933	−313.714
10	2000	403.066	−313.714	2000	−832.933	−313.714
11	1750	415.066	−313.714	4250	415.066	−313.714
12	4000	−832.933	−313.714	4000	403.066	−313.714
13	2012	−872.933	−163.714	2012	−872.933	−273.714
14	2012	403.066	−313.714	2012	403.066	−163.714
15	3988	403.066	−313.714	3988	403.066	−163.714
16	3988	−872.933	−163.714	3988	−872.933	−273.714
17	2000	−872.933	−163.714	2000	−872.933	−273.714
18	2000	403.066	−313.714	2000	403.066	−163.714
19	4000	403.066	−313.714	4000	403.066	−163.714
20	4000	−872.933	−163.714	4000	−872.933	−273.714
21	3200	−446.933	−313.714	3200	−22.933	−313.714
22	2800	−22.933	−313.714	3200	−22.933	−313.714
23	2800	−22.933	−313.714	2800	−446.933	−313.714
24	3200	−446.933	−313.714	2800	−446.933	−313.714

Appendix B. The results of Inst1 given by three methods

See [Tables B.1–B.3](#).

Data availability

No data was used for the research described in the article.

Table A.3

The information of all weld lines in instance Inst3.

Order	Start point			End point			Order	Start point			End point		
	X/mm	Y/mm	Z/mm	X/mm	Y/mm	Z/mm		X/mm	Y/mm	Z/mm	X/mm	Y/mm	Z/mm
1	0	0	490	5010	0	490	66	2840	1666	10	2840	1666	185
2	0	2516	490	5010	2516	490	67	3570	542	0	3675	542	0
3	1840	542	0	1945	542	0	68	3685	542	10	3685	542	185
4	1955	542	10	1955	542	185	69	3570	550	0	3675	550	0
5	1840	550	0	1945	550	0	70	3685	550	10	3685	550	185
6	1955	550	10	1955	550	185	71	3820	542	0	3715	542	0
7	2090	542	0	1985	542	0	72	3705	542	10	3705	542	185
8	1975	542	10	1975	542	185	73	3820	550	0	3715	550	0
9	2090	550	0	1985	550	0	74	3705	550	10	3705	550	185
10	1975	550	10	1975	550	185	75	3570	842	0	3675	842	0
11	1840	842	0	1945	842	0	76	3685	842	10	3685	842	185
12	1955	842	10	1955	842	185	77	3570	850	0	3675	850	0
13	1840	850	0	1945	850	0	78	3685	850	10	3685	850	185
14	1955	850	10	1955	850	185	79	3820	842	0	3715	842	0
15	2090	842	0	1985	842	0	80	3705	842	10	3705	842	185
16	1975	842	10	1975	842	185	81	3820	850	0	3715	850	0
17	2090	850	0	1985	850	0	82	3705	850	10	3705	850	185
18	1975	850	10	1975	850	185	83	3570	1974	0	3675	1974	0
19	1840	1974	0	1945	1974	0	84	3685	1974	10	3685	1974	185
20	1955	1974	10	1955	1974	185	85	3570	1966	0	3675	1966	0
21	1840	1966	0	1945	1966	0	86	3685	1966	10	3685	1966	185
22	1955	1966	10	1955	1966	185	87	3820	1974	0	3715	1974	0
23	2090	1974	0	1985	1974	0	88	3705	1974	10	3705	1974	185
24	1975	1974	10	1975	1974	185	89	3820	1966	0	3715	1966	0
25	2090	1966	0	1985	1966	0	90	3705	1966	10	3705	1966	185
26	1975	1966	10	1975	1966	185	91	3570	1674	0	3675	1674	0
27	1840	1674	0	1945	1674	0	92	3685	1674	10	3685	1674	185
28	1955	1674	10	1955	1674	185	93	3570	1666	0	3675	1666	0
29	1840	1666	0	1945	1666	0	94	3685	1666	10	3685	1666	185
30	1955	1666	10	1955	1666	185	95	3820	1674	0	3715	1674	0
31	2090	1674	0	1985	1674	0	96	3705	1674	10	3705	1674	185
32	1975	1674	10	1975	1674	185	97	3820	1666	0	3715	1666	0
33	2090	1666	0	1985	1666	0	98	3705	1666	10	3705	1666	185
34	1975	1666	10	1975	1666	185	99	4435	542	0	4540	542	0
35	2705	542	0	2810	542	0	100	4550	542	10	4550	542	185
36	2820	542	10	2820	542	185	101	4435	550	0	4540	550	0
37	2705	550	0	2810	550	0	102	4550	550	10	4550	550	185
38	2820	550	10	2820	550	185	103	4685	542	0	4580	542	0
39	2955	542	0	2850	542	0	104	4570	542	10	4570	542	185
40	2840	542	10	2840	542	185	105	4685	550	0	4580	550	0
41	2955	550	0	2850	550	0	106	4570	550	10	4570	550	185
42	2840	550	10	2840	550	185	107	4435	842	0	4540	842	0
43	2705	842	0	2810	842	0	108	4550	842	10	4550	842	185
44	2820	842	10	2820	842	185	109	4435	850	0	4540	850	0
45	2705	850	0	2810	850	0	110	4550	850	10	4550	850	185
46	2820	850	10	2820	850	185	111	4685	842	0	4580	842	0
47	2955	842	0	2850	842	0	112	4570	842	10	4570	842	185
48	2840	842	10	2840	842	185	113	4685	850	0	4580	850	0
49	2955	850	0	2850	850	0	114	4570	850	10	4570	850	185
50	2840	850	10	2840	850	185	115	4435	1974	0	4540	1974	0
51	2705	1974	0	2810	1974	0	116	4550	1974	10	4550	1974	185
52	2820	1974	10	2820	1974	185	117	4435	1966	0	4540	1966	0
53	2705	1966	0	2810	1966	0	118	4550	1966	10	4550	1966	185
54	2820	1966	10	2820	1966	185	119	4685	1974	0	4580	1974	0
55	2955	1974	0	2850	1974	0	120	4570	1974	10	4570	1974	185
56	2840	1974	10	2840	1974	185	121	4685	1966	0	4580	1966	0
57	2955	1966	0	2850	1966	0	122	4570	1966	10	4570	1966	185
58	2840	1966	10	2840	1966	185	123	4435	1674	0	4540	1674	0
59	2705	1674	0	2810	1674	0	124	4550	1674	10	4550	1674	185
60	2820	1674	10	2820	1674	185	125	4435	1666	0	4540	1666	0
61	2705	1666	0	2810	1666	0	126	4550	1666	10	4550	1666	185
62	2820	1666	10	2820	1666	185	127	4685	1674	0	4580	1674	0
63	2955	1674	0	2850	1674	0	128	4570	1674	10	4570	1674	185
64	2840	1674	10	2840	1674	185	129	4685	1666	0	4580	1666	0
65	2955	1666	0	2850	1666	0	130	4570	1666	10	4570	1666	185

Table B.1

The timetable of weld line in Inst1 given by the expert-tuned method.

	Manipulator 1 (k_1)			Manipulator 2 (k_2)		
	Lines ID	Start time	End time	Lines ID	Start time	End time
Region 1 ($\mathcal{L}_1$)	17	0.000	41.000	18	0.000	41.000
	14	48.782	75.449	11	51.234	77.901
	16	76.049	102.715	8	81.901	108.568
	6	117.094	143.760	15	111.313	137.980
	5	145.760	172.427	13	138.580	165.247
	4	174.427	201.094	10	169.239	195.906
	3	203.094	229.760	9	197.906	224.573
	2	231.760	258.427	12	228.573	255.239
	1	260.427	287.094	–	–	–
	7	289.094	315.760	–	–	–
Region 2 ($\mathcal{L}_2$)	35	385.760	426.760	36	385.760	426.760
	32	434.543	461.209	29	436.995	463.661
	34	461.809	488.476	26	467.661	494.328
	24	502.854	529.521	33	497.074	523.741
	23	531.521	558.188	31	524.341	551.007
	22	560.188	586.854	28	555.000	581.666
	21	588.854	615.521	27	583.666	610.333
	20	617.521	644.188	30	614.333	641.000
	19	646.188	672.854	–	–	–
	25	674.854	701.521	–	–	–
Region 3 ($\mathcal{L}_3$)	53	771.521	812.521	54	771.521	812.521
	50	820.303	846.970	47	822.755	849.422
	52	847.570	874.236	44	853.422	880.089
	42	888.615	915.281	51	882.834	909.501
	41	917.281	943.948	49	910.101	936.768
	40	945.948	972.615	46	940.760	967.427
	39	974.615	1001.281	45	969.427	996.094
	38	1003.281	1029.948	48	1000.094	1026.760
	37	1031.948	1058.615	–	–	–
	43	1060.615	1087.281	–	–	–

Table B.2

The timetable of weld line in Inst1 calculated by WAG+VNS.

	Manipulator 1 (k_1)			Manipulator 2 (k_2)		
	Lines ID	Start time	End time	Lines ID	Start time	End time
Region 1 ($\mathcal{L}_1$)	4	0.000	26.667	12	0.000	26.667
	5	28.667	55.333	9	30.667	57.333
	6	57.333	84.000	10	59.333	86.000
	3	86.000	112.667	8	88.000	114.667
	7	114.667	141.333	11	118.667	145.333
	1	143.333	170.000	16	162.916	189.582
	2	172.000	198.667	18	197.645	238.645
	17	212.021	253.021	15	246.427	273.093
	14	260.803	287.470	13	273.693	300.360
	22	370.360	397.027	30	370.360	397.027
Region 2 ($\mathcal{L}_2$)	23	399.027	425.693	27	401.027	427.693
	24	427.693	454.360	28	429.693	456.360
	21	456.360	483.027	26	458.360	485.027
	25	485.027	511.693	29	489.027	515.693
	19	513.693	540.360	34	533.276	559.942
	20	542.360	569.027	36	568.005	609.005
	35	582.381	623.381	33	616.787	643.453
	32	631.163	657.830	31	644.053	670.720
	40	740.720	767.387	48	740.720	767.387
	41	769.387	796.053	45	771.387	798.053
Region 3 ($\mathcal{L}_3$)	42	798.053	824.720	46	800.053	826.720
	39	826.720	853.387	44	828.720	855.387
	43	855.387	882.053	47	859.387	886.053
	37	884.053	910.720	52	903.636	930.302
	38	912.720	939.387	54	938.365	979.365
	53	952.741	993.741	51	987.147	1013.813
	50	1001.523	1028.190	49	1014.413	1041.080

Table B.3

The timetable of weld line in Inst1 given by HGA.

	Manipulator 1 (k_1)			Manipulator 2 (k_2)		
	Lines ID	Start time	End time	Lines ID	Start time	End time
Region 1 ($\mathcal{L}_1$)	3	0.000	26.667	8	0.000	26.667
	6	28.667	55.333	11	30.667	57.333
	14	58.079	84.746	16	59.600	86.267
	17	94.367	135.367	18	96.116	137.116
	2	148.721	175.388	15	144.898	171.565
	1	177.388	204.054	13	172.165	198.831
	7	206.054	232.721	10	202.824	229.491
	4	236.721	263.388	9	231.491	258.157
	5	265.388	292.054	12	262.157	288.824
	21	362.054	388.721	26	362.054	388.721
Region 2 ($\mathcal{L}_2$)	24	390.721	417.388	29	392.721	419.388
	32	420.134	446.800	34	421.655	448.322
	35	456.421	497.421	36	458.170	499.170
	20	510.776	537.442	33	506.952	533.619
	19	539.442	566.109	31	534.219	560.886
	25	568.109	594.776	28	564.878	591.545
	22	598.776	625.442	27	593.545	620.212
	23	627.442	654.109	30	624.212	650.878
	39	724.109	750.776	44	724.109	750.776
	42	752.776	779.442	47	754.776	781.442
Region 3 ($\mathcal{L}_3$)	50	782.188	808.855	52	783.709	810.376
	53	818.476	859.476	54	820.225	861.225
	38	872.830	899.497	51	869.007	895.674
	37	901.497	928.163	49	896.274	922.940
	43	930.163	956.830	46	926.933	953.599
	40	960.830	987.497	45	955.599	982.266
	41	989.497	1016.163	48	986.266	1012.933

References

- Arakaki, R. K., & Usberti, F. L. (2018). Hybrid genetic algorithm for the open capacitated arc routing problem. *Computers & Operations Research*, 90, 221–231.
- Chakraa, H., Guérin, F., Leclercq, E., & Lefebvre, D. (2023). Optimization techniques for multi-robot task allocation problems: Review on the state-of-the-art. *Robotics and Autonomous Systems*, Article 104492.
- Contini, A., & Farinelli, A. (2021). Coordination approaches for multi-item pickup and delivery in logistic scenarios. *Robotics and Autonomous Systems*, 146, Article 103871.
- Djenadi, A., Khanouche, M. E., & Mendil, B. (2024). A lexicographic optimization-based approach for efficient task allocation in industrial transportation multi-robot systems. *Expert Systems with Applications*, 257, Article 124998.
- Eswaran, M., kumar Inkulu, A., Tamilarasan, K., Bahubalendruni, M. R., Jaideep, R., Faris, M. S., & Jacob, N. (2024). Optimal layout planning for human robot collaborative assembly systems and visualization through immersive technologies. *Expert Systems with Applications*, 241, Article 122465.
- Gerkey, B. P., & Mataric, M. J. (2004). A formal analysis and taxonomy of task allocation in multi-robot systems. *The International Journal of Robotics Research*, 23(9), 939–954.
- Ghassemi, P., & Chowdhury, S. (2022). Multi-robot task allocation in disaster response: Addressing dynamic tasks with deadlines and robots with range and payload constraints. *Robotics and Autonomous Systems*, 147, Article 103905.
- Guo, Q., Yang, Z., Xu, J., Jiang, Y., Wang, W., Liu, Z., Zhao, W., & Sun, Y. (2024). Progress, challenges and trends on vision sensing technologies in automatic/intelligent robotic welding: State-of-the-art review. *Robotics and Computer-Integrated Manufacturing*, 89, Article 102767.
- He, P., & Hao, J.-K. (2023). Memetic search for the minmax multiple traveling salesman problem with single and multiple depots. *European Journal of Operational Research*, 307(3), 1055–1070.
- He, P., Hao, J.-K., & Wu, Q. (2021). Grouping memetic search for the colored traveling salesman problem. *Information Sciences*, 570, 689–707.
- Helsgaun, K. (2000). An effective implementation of the Lin-Kernighan traveling salesman heuristic. *European Journal of Operational Research*, 126(1), 106–130.
- Kim, H.-J., Kim, Y.-D., & Lee, D.-H. (2005). Scheduling for an arc-welding robot considering heat-caused distortion. *Journal of the Operational Research Society*, 56(1), 39–50.
- Kim, K.-Y., Kim, D.-W., & Nnaji, B. O. (2002). Robot arc welding task sequencing using genetic algorithms. *IIE Transactions*, 34, 865–880.
- Ko, D., & Han, J. (2022). A rollout heuristic algorithm for order sequencing in robotic compact storage and retrieval systems. *Expert Systems with Applications*, 203, Article 117396.
- Korsah, G. A., Stentz, A., & Dias, M. B. (2013). A comprehensive taxonomy for multi-robot task allocation. *The International Journal of Robotics Research*, 32(12), 1495–1512.

- Lee, M.-L., Behdad, S., Liang, X., & Zheng, M. (2022). Task allocation and planning for product disassembly with human-robot collaboration. *Robotics and Computer-Integrated Manufacturing*, 76, Article 102306.
- Lee, J., Kim, B.-I., & Nam, M. (2023). Novel method for welding gantry robot scheduling at shipyards. *International Journal of Production Research*, 61(17), 5842–5859.
- Liu, S., Shen, J., Tian, W., Lin, J., Li, P., & Li, B. (2023). Balanced task allocation and collision-free scheduling of multi-robot systems in large spacecraft structure manufacturing. *Robotics and Autonomous Systems*, 159, Article 104289.
- Lopes, T. C., Sikora, C. G. S., Molina, R. G., Schibelbain, D., Rodrigues, L. C. A., & Magatão, L. (2017). Balancing a robotic spot welding manufacturing line: An industrial case study. *European Journal of Operational Research*, 263(3), 1033–1048.
- Lu, Y., Benlic, U., Wu, Q., & Peng, B. (2019). Memetic algorithm for the multiple traveling repairman problem with profits. *Engineering Applications of Artificial Intelligence*, 80, 35–47.
- Martin, J. G., Muros, F. J., Maestre, J. M., & Camacho, E. F. (2023). Multi-robot task allocation clustering based on game theory. *Robotics and Autonomous Systems*, 161, Article 104314.
- Miller, C. E., Tucker, A. W., & Zemlin, R. A. (1960). Integer programming formulation of traveling salesman problems. *Journal of the ACM*, 7(4), 326–329.
- Murugappan, E., Subramanian, N., Rahman, S., Goh, M., & Chan, H. K. (2022). Performance analysis of clustering methods for balanced multi-robot task allocations. *International Journal of Production Research*, 60(14), 4576–4591.
- Mutti, S., Nicola, G., Beschi, M., Pedrocchi, N., & Tosatti, L. M. (2021). Towards optimal task positioning in multi-robot cells, using nested meta-heuristic swarm algorithms. *Robotics and Computer-Integrated Manufacturing*, 71, Article 102131.
- Qin, X., Liao, Z., Liu, C., & Xiong, Z. (2024). Online task allocation and scheduling in multi-manipulator system considering collision constraints and unknown tasks. *Robotics and Computer-Integrated Manufacturing*, 90, Article 102808.
- Qin, H., Shao, S., Wang, T., Yu, X., Jiang, Y., & Cao, Z. (2023). Review of autonomous path planning algorithms for mobile robots. *Drones*, 7(3), 211.
- Ren, J., Hao, J.-K., Wu, F., & Fu, Z.-H. (2022). Intensification-driven local search for the traveling repairman problem with profits. *Expert Systems with Applications*, 202, Article 117072.
- Rout, A., Deepak, B., Biswal, B. B., & Mahanta, G. B. (2021). Weld seam detection, finding, and setting of process parameters for varying weld gap by the utilization of laser and vision sensor in robotic arc welding. *IEEE Transactions on Industrial Electronics*, 69(1), 622–632.
- Shorinwa, O., Haksar, R. N., Washington, P., & Schwager, M. (2023). Distributed multirobot task assignment via consensus ADMM. *IEEE Transactions on Robotics*, 39, 1781–1800.
- Verma, J. K., & Ranga, V. (2021). Multi-robot coordination analysis, taxonomy, challenges and future scope. *Journal of Intelligent and Robotic Systems*, 102, 1–36.
- Wang, H., Chen, W., & Wang, J. (2020). Coupled task scheduling for heterogeneous multi-robot system of two robot types performing complex-schedule order fulfillment tasks. *Robotics and Autonomous Systems*, 131, Article 103560.
- Wang, Y., Wang, X., Chen, S., & Gu, X. (2023). Multi-station multi-robot welding system planning and scheduling based on STNSGA-D: An industrial case study. *IEEE Transactions on Automation Science and Engineering*.
- Wu, T., Zhang, Z., Zhang, Y., & Zeng, Y. (2023). Modelling and optimisation of two-sided disassembly line balancing problem with human-robot interaction constraints. *Expert Systems with Applications*, 230, Article 120589.
- Yan, F., & Di, K. (2023). Solving the multi-robot task allocation with functional tasks based on a hyper-heuristic algorithm. *Applied Soft Computing*, 146, Article 110628.
- Yan, F., Di, K., Ge, B., Liu, L., Wang, Z., Fan, W., & Hu, D. (2024). Multi-robot task allocation for optional tasks with hidden workload: Using a model-based hyper-heuristic strategy. *Engineering Applications of Artificial Intelligence*, 138, Article 109423.
- Yin, T., Zhang, Z., Zhang, Y., Wu, T., & Liang, W. (2022). Mixed-integer programming model and hybrid driving algorithm for multi-product partial disassembly line balancing problem with multi-robot workstations. *Robotics and Computer-Integrated Manufacturing*, 73, Article 102251.
- Zhao, W., Liu, Y., Wang, Y., & Yue, X. (2023). Distributed spot welding task allocation and sequential planning for multi-station multi-robot coordinate assembly processes. *International Journal of Advanced Manufacturing Technology*, 127(11–12), 5233–5251.
- Zhou, B., Zhou, R., Gan, Y., Fang, F., & Mao, Y. (2022). Multi-robot multi-station cooperative spot welding task allocation based on stepwise optimization: An industrial case study. *Robotics and Computer-Integrated Manufacturing*, 73, Article 102197.